

	ITT Manual for Buffalo				
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ITT Manual for Buffalo



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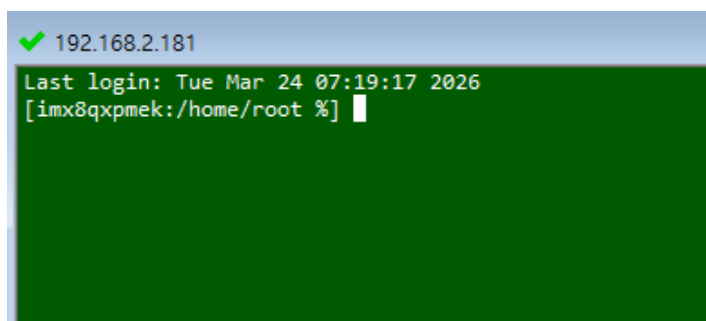
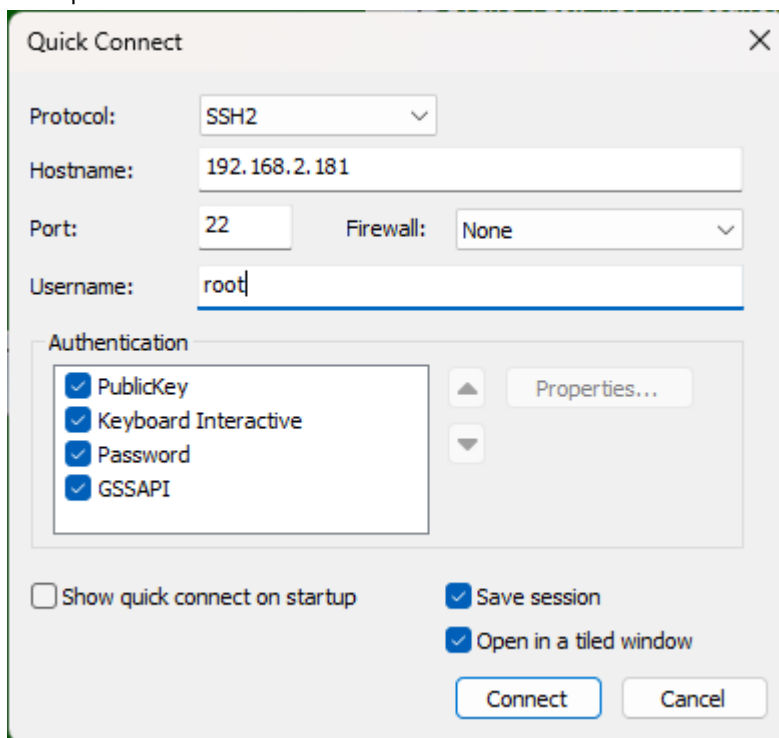
1. Connect to OBU

1.1. SSH2

Connect to OBU via SSH2 using a terminal program. Enter the OBU IP address and use root as the Username. **the initial IP address of the OBU is set to 192.168.2.181.**

the initial IP address of the RSU is set to 192.168.2.180.

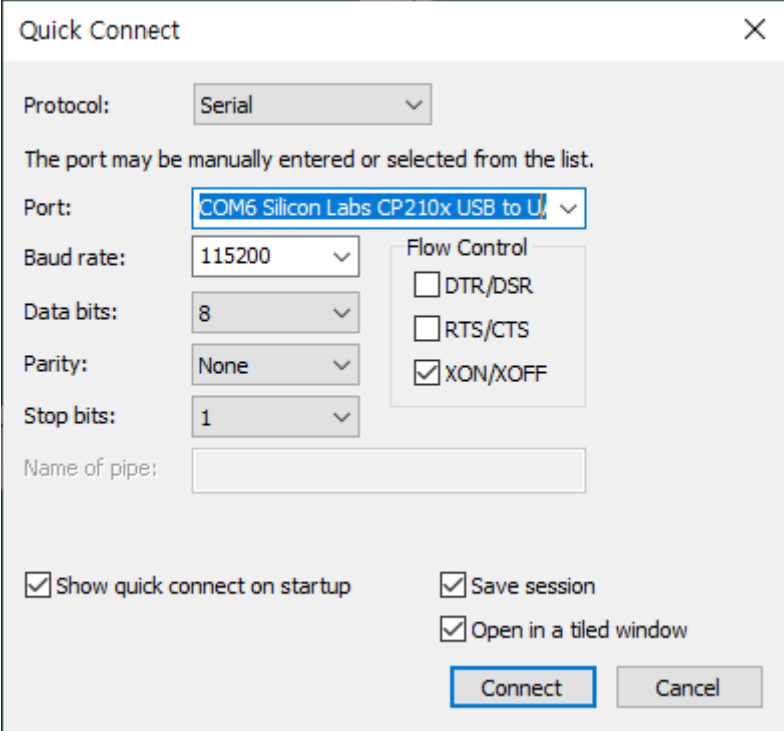
Example Connection Screen



1.2. Serial Connection

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If SSH2 connection is not possible, USB serial connection is also possible. Connect using a terminal program. To connect, select the port as USB Serial port and the remaining connection information is as follows.



The image shows a 'Quick Connect' dialog box with the following settings:

- Protocol:** Serial
- Port:** COM6 Silicon Labs CP210x USB to U (highlighted)
- Baud rate:** 115200
- Data bits:** 8
- Parity:** None
- Stop bits:** 1
- Flow Control:**
 - ☐ DTR/DSR
 - ☐ RTS/CTS
 - ☒ XON/XOFF
- Name of pipe:** (empty text field)
- Checkboxes:**
 - ☒ Show quick connect on startup
 - ☒ Save session
 - ☒ Open in a tiled window
- Buttons:** Connect (highlighted), Cancel

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2. Changing IP Address

You can see the current IP address information.

```
cat /etc/systemd/network/10-ethernet.network
```

```

[imx8qxpmeek:/home/root %] cat /etc/systemd/network/10-ethernet.network
[Match]
Name=eth*

[Network]
#DHCP=yes
Address=192.168.2.181
Gateway=192.168.2.1
DHCPServer=no

[DHCPserver]
DNS=8.8.8.8
PoolOffset=128
PoolSize=32
[imx8qxpmeek:/home/root %]

```

You can change the IP address with modify the network info file.

```
vi /etc/systemd/network/10-ethernet.network
```

```

root@imx8qxpmeek:~# vi /etc/systemd/network/10-ethernet.network

```

```

[Match]
Name=eth*

[Network]
#DHCP=yes
Address=192.168.2.181
Gateway=192.168.2.1
DHCPServer=no

```

If you need the Gateway, you can change the Gateway value.

And rebooting.

Notice. If you forget the information after changing the IP address and cannot access the IP, you may need to find or change the IP information through a serial connection.

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3. Carma forwarding application

3.1. Configuration

3.1.1. Forwarding IP and Port

The OBU and RSU can forward the received V2X message to the specific device that is described in forward.cfg. You can configure the IP and Port number for forwarding upon reception.

The forward.cfg file is the configuration file.

```
root@imx8qxpmeek:~# cat forward.cfg
# Forward Config File
# Example)
# dbglevel = 1 (1~10)
# logfile = 1 (1: yes , 0: no)
# forwardport = 5398
# forwardip = 192.168.2.140
# psid is hexadecimal. (except 0x)
# psid = 20, 21, 8003, c00305
# txpower = 20
# duration = 100 ms
# security = 1 (1: ON , 0: OFF)
dbglevel = 1
logfile = 0
forwardip = 192.168.2.140
forwardport = 5398
psid = 8005, 20, 8002, e0000017, 7f, 82, 83, 90, e0000015
txpower = 20
duration = 100
security = 0
```

Edit the file.

```
root@imx8qxpmeek:~# vi forward.cfg
```

Set the desired IP and Port. forwardip and forwardport should be same with the external computer that want to process the forwarded V2X messages.

```
forwardip = 192.168.2.140
forwardport = 5398
```

After saving the file, the modified values will take effect if you reboot or turn the power off and on again.

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3.1.2. Receiving PSID

You can configure the PSID you wish to receive. The messages which PSID is added in forward.cfg will be forwarded to forwardip:forwardport.

```
root@imx8qxpmeek:~# cat forward.cfg
# Forward Config File
# Example)
# dbglevel = 1 (1~10)
# logfile = 1 (1: yes , 0: no)
# forwardport = 5398
# forwardip = 192.168.2.140
# psid is hexadecimal. (except 0x)
# psid = 20, 21, 8003, c00305
# txpower = 20
# duration = 100 ms
# security = 1 (1: ON , 0: OFF)
dbglevel = 1
logfile = 0
forwardip = 192.168.2.140
forwardport = 5398
psid = 8005, 20, 8002, e0000017, 7f, 82, 83, 90, e0000015
txpower = 20
duration = 100
security = 0
```

Edit the file.

```
root@imx8qxpmeek:~# vi forward.cfg
```

The initial settings are as follows.

```
psid = 8005, 20, 8002, e0000017, 7f, 82, 83, 90, e0000015
```

If you only want to forward PSIDs 20 and 8005, you can modify the settings as follows.

```
forwardport = 53
psid = 8005, 20
```

After saving the file, the modified values will take effect if you reboot or turn the power off and on again.

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4. How to install BSM app

4.1. Copy the tar file into the OBU

Please copy the tar file into the OBU using SCP or WinSCP and so on.

Tar file: setbsm.tar

Target folder: /home/root/

If you use the WinSCP, use the SCP file protocol.

ID: root

No password

4.2. Extract the tar file

After connecting SSH , please type the command.

tar -xvf setbsm.tar

```
[imx8qxpme@:/home/root %] tar -xvf setbsm.tar
```

4.3. Run script

Next, you need to give a permission like this

chmod 755 setbsm.sh

```
[imx8qxpme@:/home/root %] chmod 755 ./setbsm.sh
```

And execute it.

./setbsm.sh

```
[imx8qxpme@:/home/root %] ./setbsm.sh
```

Then reboot.

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5. How to use Python library

5.1. Source codes

You can download source code from the next link:

- Github : https://github.com/cogaug/v2x_intf_lib

Python library is provided as source code includes:

- v2xintf.py : V2X OBU/RSU interface python library
- v2xintf_ex.py : Demo python application to display received message

v2xinterface.py does not use any 3rd party library or package to import. For your application, simply import V2XInterface from v2xintf.py as shown in demo program.

5.2. Prepare V2X device

1. Setup OBU or RSU and connect to the network.
2. Set OBU or RSU IP address (2. Changing IP address)
3. Modify forwardip and forwardport in forward.cfg (3. Carma forwarding application)
 - A. The python application to receive the message should be running on the computer which IP address is forwardip.

5.3. Execute demo application

1. Require python : > 3.11
2. Run demo application

```
python v2xintf_ex.py
```

3. Demo application binds UDP socket to the port 5389 as default.
4. If you change the forwardport in forward.cfg file in OBU or RSU (3. Carma forwarding application), you have to specify binding port

```
python v2xintf_ex.py --local-port xxxx
```

5.4. Integrate library

To integrate library: (Refer the v2xintf_ex.py source code)

1. Import V2XInterface class from v2xintf.py

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```
from v2xintf import V2XInterface, WAVE_MSG_IDS
```

- WAVE_MSG_IDS includes all the information forwarded from OBU or RSU

2. Create instance of V2XInterface

```
v2x_interface = V2XInterface(callback=test_callback, local_port=xxxx)
```

- **callback** (Mandatory) : Call back method to be invoked when the message received.
- **local_port**(Optional) : local port to receive message
- **remote_address** and **remote_port** is for sending message (Not tested yet, To be added)

3. Call start method

```
v2x_interface.start()
```

4. Enter infinite loop to keep the application.

```
try:
    while True:
        time.sleep(0.1)
except KeyboardInterrupt:
    v2x_interface.stop()
```